$\therefore y(4) = 4^3 - 2 \cdot 4^2 + 1$

$= 64 - 32 + 1$

$= 33$

(Ans:)

Problem: Derive Newton's General interpolation formula.

or,

Derive Newton's divided difference interpolation formula for unequal interval?

Sol^n: Let $f(x_0), f(x_1), \text{---} f(x_n)$ be the values of $f(x)$ corresponding to the arguments $x_0, x_1, x_2, \text{---} x_n$ not necessarily equally spaced. From the definition of divided differences,

$$f(x, x_0) = \frac{f(x) - f(x_0)}{x - x_0}$$

$$\text{or, } f(x) = f(x_0) + (x - x_0) f(x, x_0) \quad \text{---} (1)$$

Also $$f(x, x_0, x_1) = \frac{f(x, x_0) - f(x_0, x_1)}{x - x_1}$$

$$\Rightarrow f(x, x_0) = f(x_0, x_1) + (x - x_1) f(x, x_0, x_1) \quad \text{---} (2)$$

$$f(x) = f(x_0) + (x-x_0)\, f(x, x_0).$$

Sol^n: Let $y_0, y_1, \ldots y_n$ be the values of y corresponding to the arguments $x_0, x_1, \ldots x_n$ not necessarily equally spaced. From the definition of divided differences,

$$\delta(x, x_0) = \frac{y - y_0}{x - x_0}$$

$$\therefore y = y_0 + (x - x_0)\, \delta(x, x_0) \quad \text{——— (1)}$$

Also

$$\delta(x,x_0,x_1) = \frac{\delta(x,x_0)-\delta(x_0,x_1)}{x-x_1}$$

$$\Rightarrow \delta(x,x_0) = \delta(x_0,x_1) + (x-x_1)\,\delta(x,x_0,x_1) \quad \text{--- (11)}$$

similarly

$$\delta(x,x_0,x_1) = \delta(x_0,x_1,x_2) + (x-x_2)\,\delta(x,x_0,x_1,x_2)$$

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$$\delta(x,x_0,x_1,\ldots,x_{n-1}) = \delta(x_0,x_1,\ldots,x_n) + (x-x_n)\,\delta(x,x_0,x_1,\ldots,x_n)$$

putting (11) in (1), we get

$$y = y_0 + (x-x_0)\left[\delta(x_0,x_1) + (x-x_1)\,\delta(x,x_0,x_1)\right]$$

$$= y_0 + (x-x_0)\,\delta(x_0,x_1) + (x-x_0)(x-x_1)\,\delta(x,x_0,x_1)$$

$$= y_0 + (x-x_0)\,\delta(x_0,x_1) + (x-x_0)(x-x_1)\left[\delta(x_0,x_1,x_2) + (x-x_2)\,\delta(x,x_0,x_1,x_2)\right] \quad \text{From (11)}$$

$$= y_0 + (x-x_0)\,\delta(x_0,x_1) + (x-x_0)(x-x_1)\,\delta(x_0,x_1,x_2) + (x-x_0)(x-x_1)(x-x_2)\,\delta(x,x_0,x_1,x_2)$$

$$= y_0 + (x-x_0)\,\delta(x_0,x_1) + (x-x_0)(x-x_1)\,\delta(x_0,x_1,x_2) + \ldots + (x-x_0)(x-x_1)\ldots(x-x_{n-1})\,\delta(x_0,x_1,\ldots,x_n) + (x-x_n)\,\delta(x,x_0,\ldots,x_n)$$

This formula is called Newton's divided difference/ general interpolation formula.

Problem: Find a cubic polynomial from the following data

x	y
~~-1~~	~~3~~
~~0~~	~~-6~~
~~3~~	~~39~~
~~6~~	~~822~~
~~7~~	~~1611~~

x	y
0	3
2	11
3	30
5	128

$\frac{11-3}{2-0} = \frac{8}{2}$

$\frac{30-11}{3-0}$

9

=

Soln: The divided difference table is

x	y	1st diff ~~$\delta(x,x_0)$~~	2nd diff ~~$\delta(x,x_0,x_1)$~~	3rd ~~$\delta(x,x_0,x_1,x_2)$~~
0	3			
2	11	4	5	
3	30	19	10	1
5	128	49		

By Newton's general interpolation

$\cancel{y = y_0 + (x-x_0)\,\delta(x_1,x_0) + (x-x_0)(x-x_1)\,\delta(x_0,x_1,x_2)}$

$y = y_0 + (x-x_0)\ \text{1st diff} + (x-x_0)(x-x_1)\ \text{2nd diff} + (x-x_0)(x-x_1)(x-x_2)\ \text{3rd diff}$

$= 3 + (x-0)\cdot 4 + (x-0)(x-2)\,5 + (x-0)(x-2)(x-3)\cdot 1$

$$y = 3 + 4x + x(x-2)\cdot 5 + x(x-2)(x-3)$$
$$= 3 + 4x + 5x(x-2) + (x^2-2x)(x-3)$$
$$= 3 + 4x + 5x^2 - 10x + (x^3 - 3x^2 - 2x^2 + 6x)$$
$$= 3 - \cancel{6x} + \cancel{5x^2} + x^3 - \cancel{5x^2} + \cancel{6x}$$
$$= x^3 + 3$$

(Ans:)

2014 **Example 9.23:** Using the following table find $f(x)$ as a polynomial in x.

x	$f(x)$
-1	3
0	-6
3	39
6	822
7	1611

$\frac{39+6}{3-0} = \frac{45}{3}$

$\frac{822-39}{6-3} = 261$

Sol^n: The divided difference table is

x	$y=f(x)$				
-1	3				
		-9			
0	-6		6		
		15		5	
3	39		41		1
		261		13	
6	822		132		
		789			
7	1611				

By Newton's general interpolation

$y = y_0 + (x-x_0)\,\text{1st diff} + (x-x_0)(x-x_1)\,\text{2nd diff.} + (x-x_0)(x-x_1)(x-x_2)\,\text{3rd diff.} + (x-x_0)(x-x_1)(x-x_2)(x-x_3)\,\text{4th diff}$

$= 3 + (x+1)\cdot(-9) + (x+1)(x-0)\cdot 6 + (x+1)(x-0)(x-3)5 + (x+1)(x-0)(x-3)(x-6)\cdot 1$

$= 3 - 9x - 9 + x(x+1)6 + x(x+1)(x-3)5 + x(x+1)(x^2-6x-3x+18)$

$= -9x - 6 + 6x^2 + 6x + 5x(x^2-3x+x-3) + (x^2+x)(x^2-9x+18)$

$= 6x^2 - 3x - 6 + 5x(x^2-2x-3) + x^4 - 9x^3 + 18x^2 + x^3 - 9x^2 + 18x$

$= x^4 - 8x^3 + 24x^2 - 9x^2 - 3x + 18x - 6 + 5x^3 - 10x^2 - 15x$

$= x^4 - 3x^3 + 24x^2 - 19x^2 \cancel{-18x} \cancel{+18x} - 6$

$= x^4 - 3x^3 + 5x^2 - 6$

(Ans:)

Problem: Certain corresponding values of x and $\log_{10} x$ are $(300, 2.4771)$, $(304, 2.4829)$, $(305, 2.4843)$ and $(307, 2.4871)$. Find $\log_{10} 301$ by using Newton general formula / Newton divided difference formula.

$\delta(x_0, x_1, x_2)$

$\delta(x_1, x_2, x_3)$

Sol^n: The divided difference table is

x	$y = \log_{10} x$	1st diff.	2nd diff.	3rd diff
300	2.4771			
		0.00145		
304	2.4829		-0.00001	
		0.00140		
305	2.4843		0	
		0.00140		
307	2.4871			

By Newton's general interpolation

$$y(\log_{10} 301) = \log_{10} 301 = y_0 + (x - x_0)\underbrace{\text{1st diff}}_{\delta(x_0, x_1)} + (x-x_0)(x-x_1)\underbrace{\text{2nd diff}}_{\delta(x_0, x_1, x_2)} + \dots$$

$$= 2.4771 + (301-300)(0.00145) + (301-300)(301-304)(-0.00001)$$

$$= 2.4771 + 0.00145 + 0.00003$$

$$= 2.4786$$

(Ans.)

2015, 2013, 2011

Problem: Derive Lagrange's interpolation (inverse interpolation) formula?

sol^n:

Problem: Find a polynomial of x and find $f(2)$.

x	y
0	-1
3	11
4	35
6	144

$\frac{35-11}{4-3} = \frac{24}{3}$ 8

$\frac{144-35}{2}$

9

sol^n: The divided difference table is.

x	y	1st diff	2nd diff	3rd diff
0	-1			
		4		
3	11		5	1
4	35	24	11	
6	149	57		

$\delta(x_0, x_1, x_2)$

$\delta(x_1, x_2, x_3)$

$\delta(x_0, x_1, x_2, x_3)$

By newton's general interpolation formula is

$$y = y_0 + (x-x_0)\ \text{1st diff} + (x-x_0)(x-x_1)\ \text{2nd diff} + (x-x_0)(x-x_1)(x-x_2)\ \text{3rd diff}$$

$$= -1 + (x-0)\,4 + (x-0)(x-3)\,5 + (x-0)(x-3)(x-4)\cdot 1$$

$$= -1 + 4x + x(5x-15) + x(x^2-4x-3x+12)$$

$$y = -1 + 4x + 5x^2 - 15x + x(x^2 - 7x + 12)$$
$$= 5x^2 - 11x - 1 + x^3 - 7x^2 + 12x$$
$$= x^3 - 2x^2 + x - 1$$

and obtain $x = 2$

$$f(2) = (2)^3 - 2.2^2 + 2 - 1$$
$$= 8 - 2.4 + 1$$
$$= 1$$

(Ans.)

2015, 2013, 2011

Sec-B **Question:** Derive Lagrange's interpolation (inverse interpolation) formula?

Sol: Let $f(x)$ denote a polynomial of the nth degree which takes the values $y_0, y_1, y_2, \ldots, y_n$ when x has the $x_0, x_1, \ldots, x_n$ respectively. We know nth difference of a polynomial of degree n is constant and hence $(n+1)$ difference of polynomial of degree n is zero

From the defn of divided difference

$$\delta(x_1, x_0) = \delta(x_0, x_1) = \frac{y_0 - y_1}{x_0 - x_1} = \frac{y_1 - y_0}{x_1 - x_0} = \frac{y_1}{x_1 - x_0} + \frac{y_0}{x_0 - x_1}$$

$$\delta(x_2, x_1, x_0) = \frac{\delta(x_2, x_1) - \delta(x_1, x_0)}{x_2 - x_0} = \frac{1}{x_2 - x_0}\left[\frac{y_2 - y_1}{x_2 - x_1} - \frac{y_1 - y_0}{x_1 - x_0}\right]$$

$$= \frac{1}{(x_2-x_0)}\left[\frac{y_2}{x_2-x_1} - \frac{y_1}{x_2-x_1} - \frac{y_1}{x_1-x_0} + \frac{y_0}{x_1-x_0}\right]$$

$$= \frac{1}{(x_2-x_0)}\left[\frac{y_2}{x_2-x_1} + \frac{y_1}{(x_1-x_2)} + \frac{y_1}{x_1-x_0} - \frac{y_0}{x_0-x_1}\right]$$

$$= \frac{1}{(x_2-x_0)}\left[\frac{y_2}{x_2-x_1} + \frac{y_1(\cancel{x_1}-x_0-\cancel{x_1}+x_2)}{(x_1-x_2)(x_1-x_0)} - \frac{y_0}{(x_0-x_1)}\right]$$

$$= \frac{y_2}{(x_2-x_0)(x_2-x_1)} + \frac{y_1\cancel{(x_2-x_0)}}{(x_1-x_2)(x_1-x_0)\cancel{(x_2-x_0)}} + \frac{y_0}{(x_0-x_1)(x_0-x_2)}$$

$$\delta(x_2,x_1,x_0) = \frac{y_2}{(x_2-x_0)(x_2-x_1)} + \frac{y_1}{(x_1-x_0)(x_1-x_2)} + \frac{y_0}{(x_0-x_1)(x_0-x_2)}$$

$$\therefore \delta(x_0,x_1,x_2,\dots x_n) = \frac{y_0}{(x_0-x_1)(x_0-x_2)\dots(x_0-x_n)} + \frac{y_1}{(x_1-x_0)(x_1-x_2)\dots(x_1-x_n)} + \dots + \frac{y_n}{(x_n-x_0)(x_n-x_1)\dots(x_n-x_{n-1})} \quad \text{--- (1)}$$

From equation (1) $(n+1)$th difference of a polynomial of degree n is zero i.e

$$\delta(x_0,x_0,x_1,\dots,x_n) = 0$$

$$\frac{y}{(x-x_0)(x-x_1)\cdots(x-x_n)} + \frac{y_0}{(x_0-x)(x_0-x_1)(x_0-x_2)\cdots(x_0-x_n)} + \cdots + \frac{y_n}{(x_n-x)(x_n-x_0)\cdots(x_n-x_{n-1})} = 0$$

$$\Rightarrow \frac{y}{(x-x_0)(x-x_1)\cdots(x-x_n)} = \frac{y_0}{(x-x_0)(x_0-x_1)(x_0-x_2)\cdots(x_0-x_n)} + \frac{y_1}{(x-x_1)(x_1-x_0)\cdots(x_1-x_n)} + \cdots + \frac{y_n}{(x-x_n)(x_n-x_0)\cdots(x_n-x_{n-1})}$$

$$\Rightarrow y = \frac{\cancel{(x-x_0)}(x-x_1)\cdots(x-x_n)}{\cancel{(x-x_0)}(x_0-x_1)\cdots(x_0-x_n)}y_0 + \frac{(x-x_0)\cancel{(x-x_1)}\cdots(x-x_n)}{\cancel{(x-x_1)}(x_1-x_0)\cdots(x_1-x_n)}y_1 + \cdots + \frac{(x-x_0)(x-x_1)\cdots\cancel{(x-x_n)}}{\cancel{(x-x_n)}(x_n-x_0)\cdots(x_n-x_{n-1})}y_n$$

$$= \frac{(x-x_1)\cdots(x-x_n)}{(x_0-x_1)\cdots(x_0-x_n)}y_0 + \frac{(x-x_0)(x-x_2)\cdots(x-x_n)}{(x_1-x_0)\cdots(x_1-x_n)}y_1 + \cdots + \frac{(x-x_0)(x-x_1)\cdots(x-x_{n-1})}{(x_n-x_0)\cdots(x_n-x_{n-1})}y_n$$

This is lagranges interpolation formula.